

Integrating Unsupervised Chemical Space Mapping with Supervised Learning for hERG Cardiotoxicity Prediction

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Background

There are several protein targets in drug discovery which can be used as indicators of drug toxicity, and all drugs need to be tested against these targets to prove their safety. One of these targets is the voltage gated inwardly rectifying potassium channel KCNH2 or hERG (human ether -a-go-go -related gene). This is the alpha subunit of a potassium ion channel. It controls the repolarization of the current in cardiac action potential and coordinates the heart's beating. If this channels activity is inhibited, it can lead to fatal disorders such as long QT syndrome which can lead to irregular heartbeats, fainting, seizures or death. The hERG channel has a large cavity in the conducting pore which can accommodate various drug like molecules which can inhibit the activity. Residues such as tyrosine and phenylalanine are exposed which can form strong hydrophobic interactions with various drug like molecules. Various drugs have the potential to block this channel and have fatal effects. Therefore, predicting the effect on hERG is an important step in a drug discovery pipeline to filter out toxic drug candidates. Unsupervised learning through PCA and UMAP can allow us to project drug candidates onto a chemical space and cluster them through structural similarity. It can help us identify chemical structures strongly associated with inhibition of hERG . Information about chemical from the unsupervised machine learning could potentially inform and improve the performance of supervised machine learning models in accurately predicting the effect of candidate drug molecules on hERG activity

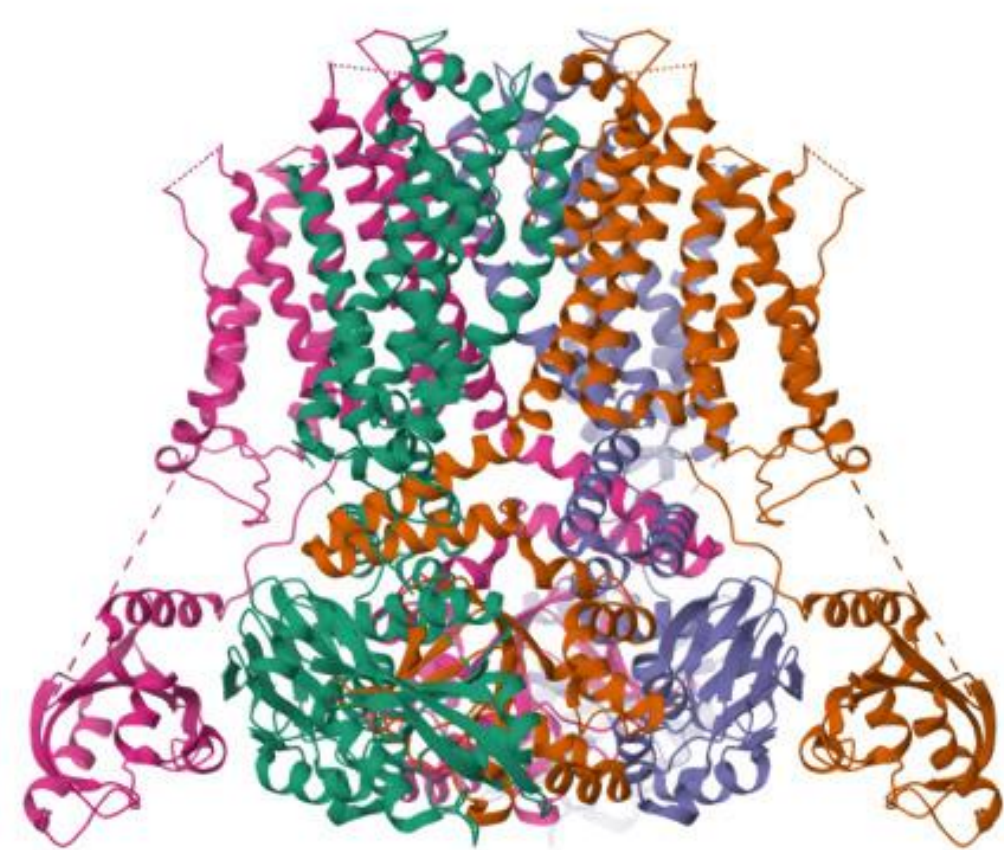


Figure 1: hERG 3D structure from PDB

Hypothesis

Chemical structures strongly associated with hERG toxicity can be identified from the chemical space structure through unsupervised machine learning. Knowledge of these chemical structures can significantly improve supervised machine learning models which predict hERG toxicity.

Computational Methods

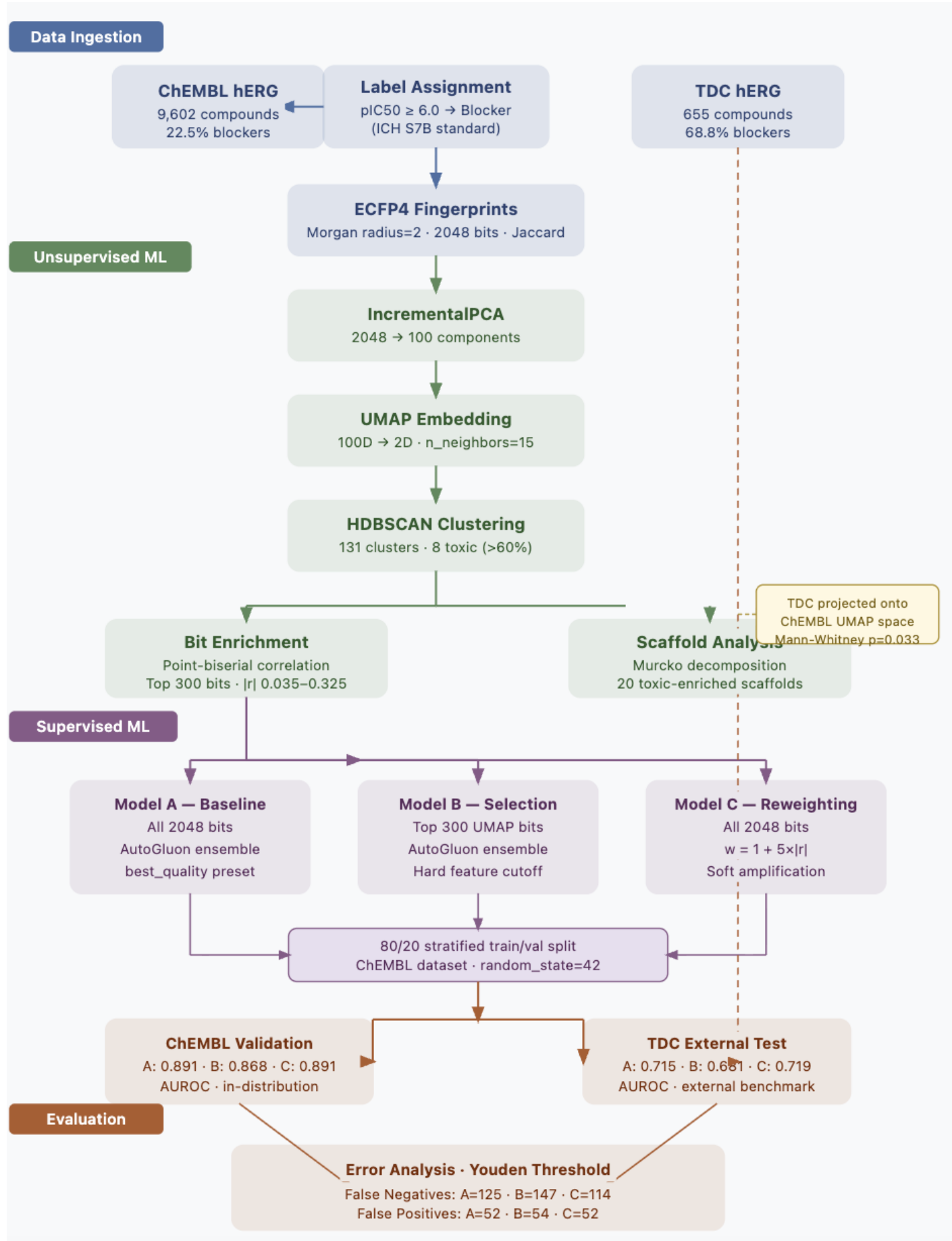


Figure 2. Overall computational design

UMAP Projection from unsupervised ML

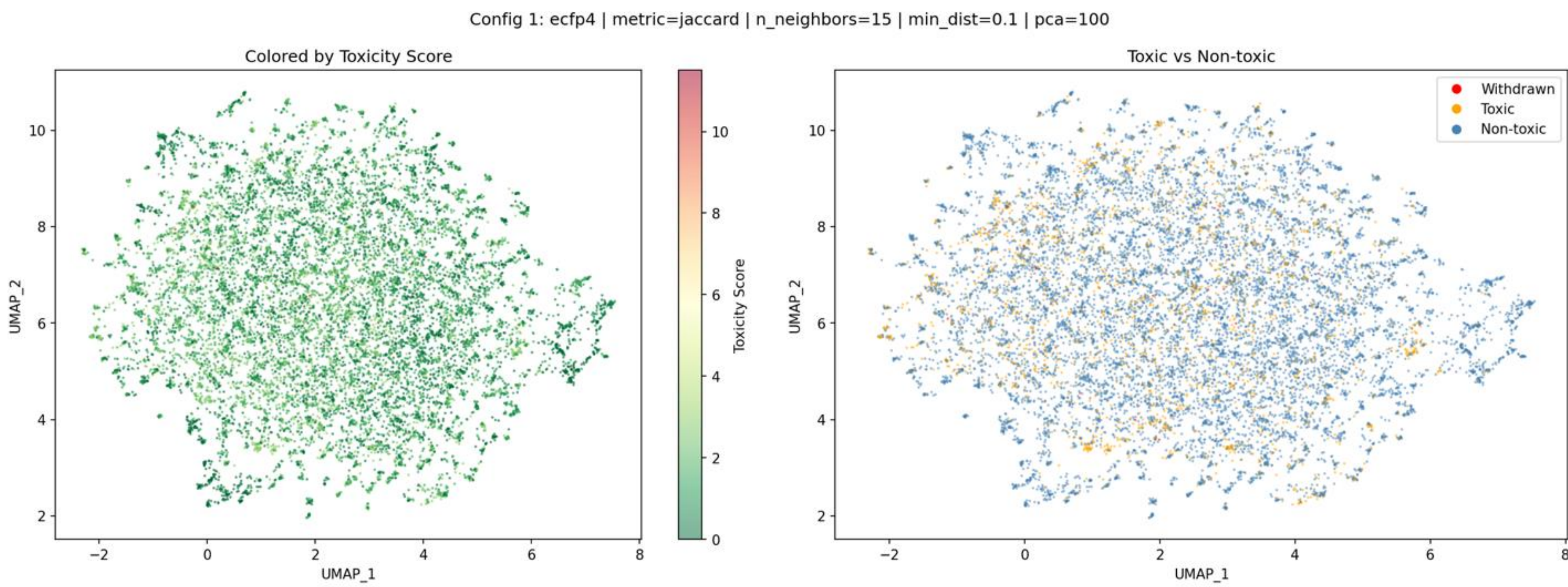


Figure 3. 2D UMAP Projection of the chemical space of ChEMBL data from unsupervised ML. The resulting embedding revealed a continuous chemical space with small clusters interspaced. There is no clear separation between toxic and non-toxic groups. HDBSCAN clustering (min_cluster_size=30, min_samples=10) identified 131 stable clusters from this embedding, with 1,964 compounds (20.5%) assigned as noise. 8 (6.1%) were designated as toxic clusters (clusters where > 60% of member compounds are hERG blockers (pIC50 ≥ 6.0)). The remaining 123 clusters had blocker rates below this threshold, with a mean blocker rate of 0.231 across all clusters. The maximum blocker rate observed in any single cluster was 0.885.

Enriched ECFP bits and Murcko Scaffolds

Enriched ECFP bit	Absolute Correlation
454	0.3249
38	0.294802
315	0.287784
1338	0.259297
88	0.248337
984	0.240168
293	0.237200
450	0.214932
1314	0.213646

Figure 4: Top 10 bits positively correlated with toxicity. Point-biserial correlation was computed between each of the 2,048 ECFP4 bits and the binary in_toxic_region label derived from HDBSCAN cluster assignments. The top 300 bits by absolute correlation were selected for downstream use in Models B and C. Correlations ranged from 0.035 to 0.325, with a mean of 0.070 across the selected bits. Bit 454 showed the strongest association ($|r| = 0.325$), followed by bits 38, 315, and 1338. Notably, all retained correlations were positive, indicating that the presence of specific substructural features (no absences) is associated with toxicity.

scaffold	count	n_toxic	mean_pIC50	toxic_rate
<chem>c1ccc(OC2C63CCC(C2)N3Cc2cccs2)cc1</chem>	5	5	8.5770	1.0
<chem>O=c1cc(OCc2ccccc2)ccn1-c1ccc(OCCN2CCCC2)cc1</chem>	8	8	8.5091	1.0
<chem>O=C(COc1ccc(C(=O)c2ccccc2)cc1)Nc1ccccc1</chem>	5	5	7.6301	1.0
<chem>c1cnc2c(OC3CCNCC3)cccc2c1</chem>	18	17	6.8850	0.94
<chem>c1ccc(-n2cc(Cn3ccnc3)nn2)cc1</chem>	69	64	7.8041	0.93

Figure 5: Highly Correlated Murcko Scaffolds. Murcko scaffolds were computed to identify the core structures strongly associated with hERG toxicity. Scaffolds represented by at least 5 compounds were ranked by toxic_rate and secondarily by mean_pIC50. Three scaffolds achieved a 100% blocker rate. All these scaffolds have a basic nitrogen capable of interacting with Tyrosine and Aspartate in the channel pore, flanked by hydrophobic aromatic groups that occupy the inner cavity which can interact with tyrosine and phenylalanine.

Supervised ML Evaluation

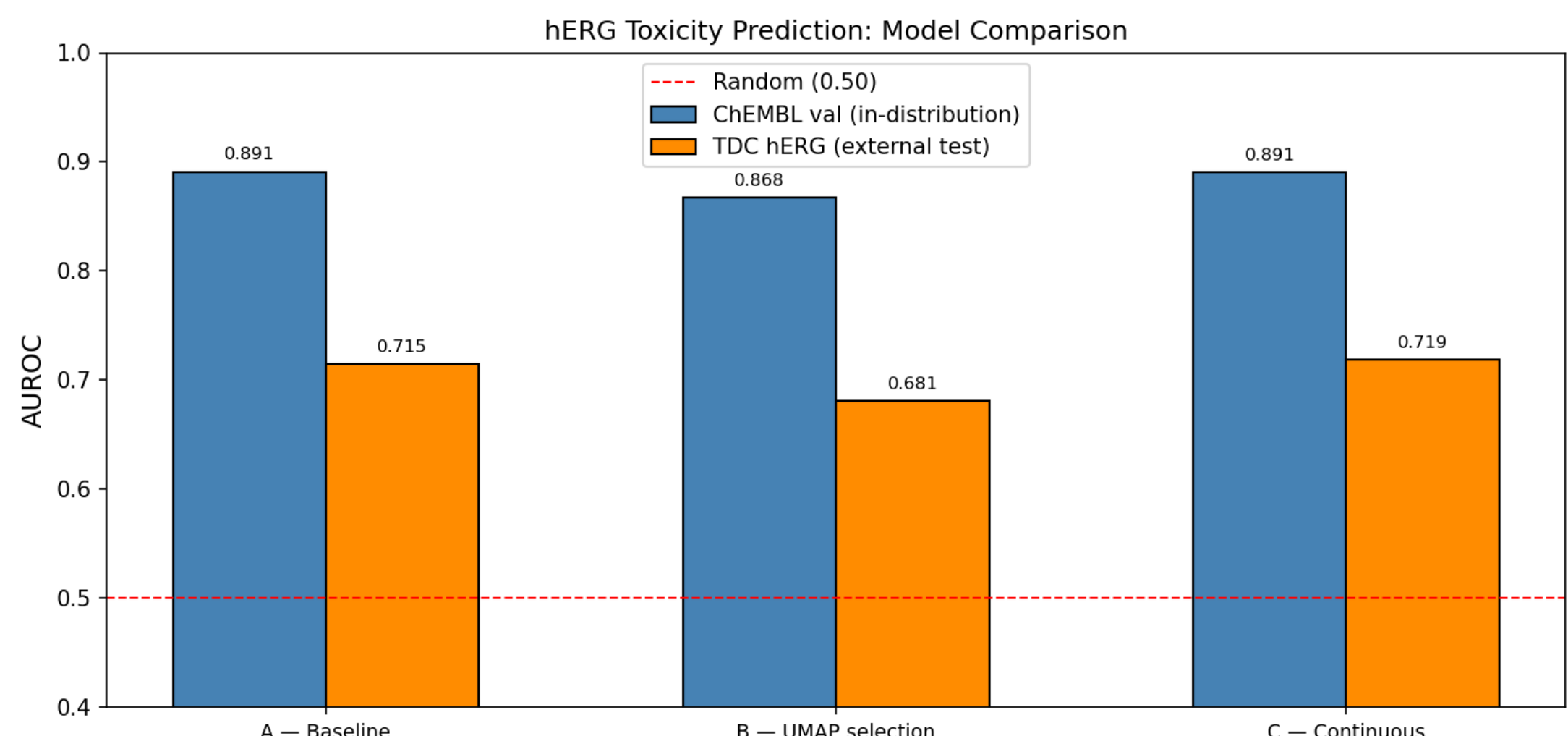


Figure 6: Supervised ML Model comparison on validation and test datasets. Model A achieved a validation AUROC of 0.891 and an external TDC AUROC of 0.715. Model B achieved a validation AUROC of 0.868 and TDC AUROC of 0.681. The reduction relative to Model A suggests that the other still contribute meaningful information. Model C has the same baseline validation AUROC of 0.891 and the highest external TDC AUROC of 0.719 which is only marginally higher than Models A and B on the held-out test set.

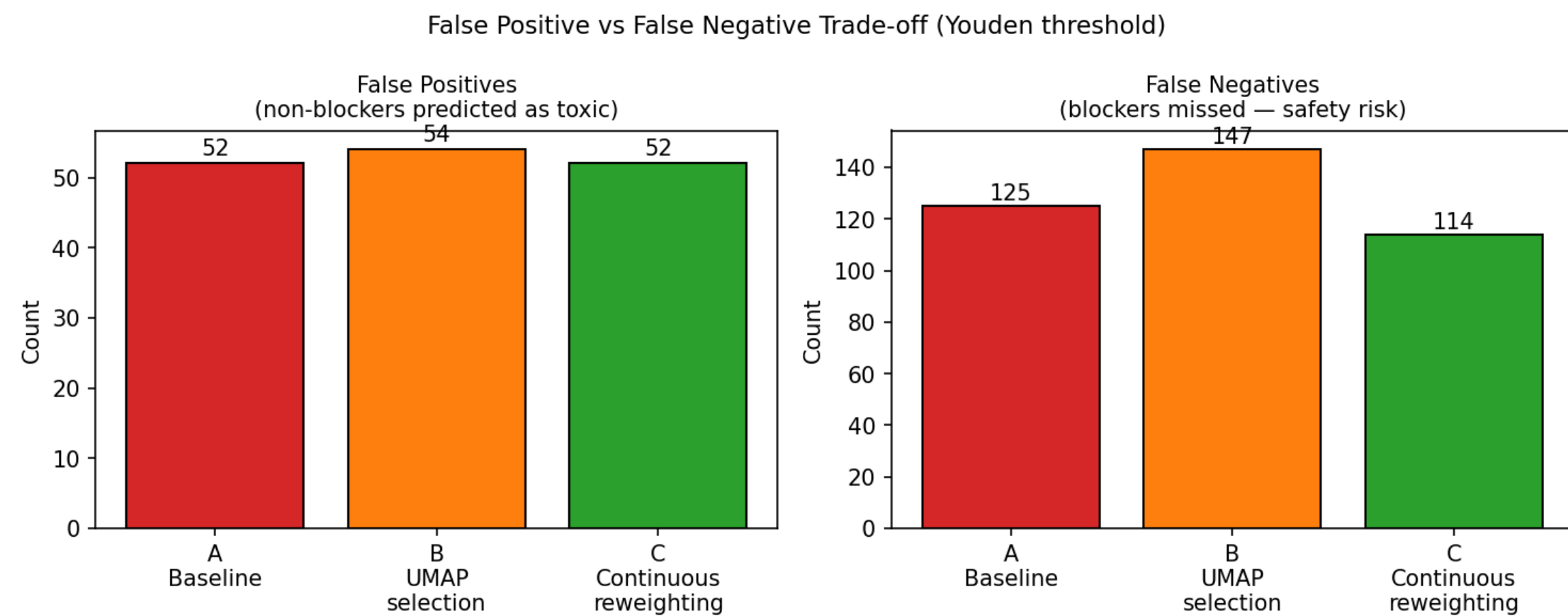


Figure 7: False positive and False negative counts between models. False positives were similar across models. False negative counts shows more variation. Model B's increase in false negatives relative to Model A is consistent with its lower TDC test score. Model C achieved the lowest false negative count of 114. Model C's reduction in false negatives along with its marginal TDC AUROC gain might make it the preferred model from a safety perspective, even though its overall discrimination is comparable to Model A.

Conclusions and Future Directions

By embedding 9,602 ChEMBL hERG compounds using UMAP and clustering with HDBSCAN, 131 clusters were identified. 8 were labeled as toxic. This analysis recovered basic nitrogen centers flanked by hydrophobic aromatic systems. Point-biserial correlation identified 300 structurally informative bits spanning a correlation range of 0.035 to 0.325, which were used to inform two feature-engineering strategies for downstream classification. Of the three AutoGluon ensemble models evaluated, the continuous reweighting approach (Model C) achieved the best balance of performance across metrics: matching the baseline validation AUROC of 0.891, achieving the highest external TDC AUROC of 0.719, and producing the lowest false negative count of 114. The projection of TDC compounds onto the ChEMBL UMAP space further confirmed that hERG blockers occupy structurally consistent chemical neighborhoods across independent datasets (Mann-Whitney $p = 0.033$), supporting the generalizability of the learned chemical space representation. Together, these results demonstrate that unsupervised structural analysis can meaningfully inform supervised toxicity prediction, soft feature amplification guided by chemical space geometry is a more effective strategy than hard feature selection.